








Time Series Analysis

Seminar 15: Review and exam preparation



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Acronyms used in this material (1/2)

- **ACF**: Autocorrelation Function
- **ADF**: Augmented Dickey–Fuller (test)
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AR**: AutoRegressive (model)
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **ARCH-LM**: ARCH Lagrange Multiplier (test)
- **ARFIMA**: AutoRegressive Fractionally Integrated Moving Average (model)
- **ARIMA**: AutoRegressive Integrated Moving Average (model)
- **ARMA**: AutoRegressive Moving Average (model)
- **BET**: Bucharest Exchange Trading (index)
- **BIC**: Bayesian Information Criterion
- **DM**: Diebold–Mariano (test)
- **ECM**: Error Correction Model
- **EODHD**: EOD Historical Data (market-data provider)
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GDP**: Gross Domestic Product
- **HICP**: Harmonised Index of Consumer Prices

Acronyms used in this material (2/2)

- **HLN**: Harvey–Leybourne–Newbold (small-sample correction of the DM test)
- **KPSS**: Kwiatkowski–Phillips–Schmidt–Shin (test)
- **LB**: Ljung–Box (test)
- **LM**: Lagrange Multiplier
- **MA**: Moving Average
- **MAE**: Mean Absolute Error
- **MASE**: Mean Absolute Scaled Error
- **OLS**: Ordinary Least Squares
- **PACF**: Partial AutoCorrelation Function
- **RMSE**: Root Mean Squared Error
- **ROBOR**: Romanian Interbank Offer Rate
- **RSS**: Residual Sum of Squares
- **SAR**: Seasonal AutoRegressive term (EViews)
- **SARIMA**: Seasonal ARIMA (model)
- **SE**: Standard Error
- **SES**: Simple Exponential Smoothing
- **SIGMASQ**: SIGMA SQuared, the innovation variance (EViews)
- **VAR**: Vector AutoRegression
- **VAT**: Value Added Tax
- **VECM**: Vector Error Correction Model

Today's Question and Seminar Structure (1/2)

- The question of the seminar
  - **can you solve, on paper and on data, an exam problem from any core chapter?**
- Results to obtain (nothing is handed in)
  - the answers to the A exercises
  - a figure and a table for each B exercise
  - the corrected AI answer of C2
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- The exercises, their types and their order: the exercise map (next slide)
- Seminar: prof. Daniel Traian Pele, danpele@ase.ro
- **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

Today's Question and Seminar Structure (2/2)

Task	Topic	Type	Chapters
A1	an ARIMA(0,1,1) by hand: moments, ACF, forecasts, exponential smoothing	Solved	0, 1, 3
A2	identifying the Romanian unemployment rate: ADF, KPSS, correlogram, Ljung–Box	Proposed	1–3
A3	SARIMA output for Romanian GDP and a forecast by hand	Proposed	4
A4	VAR output, Granger F from two RSS, a long-run effect	Proposed	6
A5	GARCH(1,1)-t output for the DAX and VaR 1%	Proposed	5
B1	Box–Jenkins for Romanian industrial production	Solved	3, 4
B2	volatility of the BET since 2015	Proposed	5
B3	the Romanian and German 10-year yields: cointegration	Proposed	3, 7
C1, C2	a forecast competition for inflation; audit an AI answer	Proposed	0–10

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in the seminar
- Part A: exam problems on paper (a calculator is enough); Part B: real data, solved in the notebook; Part C: an open question and the critique of an AI answer

Data used

Series	Source	Frequency	Period
unemployment rate, Romania (seasonally adjusted)	Eurostat (une_rt_m)	monthly	2005–2026
real GDP, Romania (not adjusted)	Eurostat (namq_10_gdp)	quarterly	2000–2026
HICP inflation and ROBOR 3M, Romania	Eurostat	monthly	2005–2026
industrial production, Romania (calendar adjusted)	Eurostat (sts_inpr_m)	monthly	2005–2026
DAX, BET	EODHD	daily	2000–2026
10-year yields, Romania and Germany	EODHD	monthly means of daily data	2008–2026

- Part A needs only a calculator: the output tables are on the slides; Part B uses the notebook
- Growth rates in %: $100\Delta \ln y_t$; annual rates: $100\Delta_{12} \ln y_t$ (monthly), $100\Delta_4 \ln y_t$ (quarterly); returns: $100\Delta \ln P_t$

Prerequisites for Today (1/22): description, stationarity, ARMA

Quantity	Formula
SES, MASE	$\ell_t = \alpha y_t + (1-\alpha)\ell_{t-1}$; MASE = MAE / MAE (seasonal naive, in sample)
ACF, band	$\hat{\rho}(h) = \hat{\gamma}(h)/\hat{\gamma}(0)$, $\pm 1.96/\sqrt{T}$
Ljung–Box	$Q^*(m) = T(T+2)\sum_{h=1}^{m} \hat{\rho}(h)^2/(T-h) \sim \chi^2(m-k)$, k = ARMA parameters
AR(1), MA(1)	$\rho(h) = \phi^h$; $\rho(1) = \theta/(1+\theta^2)$, $\rho(h) = 0$ for $h \geq 2$
Identification	ACF cuts off: MA; PACF cuts off: AR; both decay: ARMA
Criteria	AIC $= -2\ln L + 2k$, BIC $= -2\ln L + k \ln T$; the smallest wins
Forecast interval	$\hat{y}_{T+h} \pm 1.96\,\sigma\sqrt{\sum_{j<h}\psi_j^2}$

Prerequisites for Today (2/22): the notation of the previous sheet

- Smoothing and accuracy
  - ℓ_t: the smoothed level; $\alpha \in (0, 1)$: the weight of the newest observation y_t
  - MAE: the mean absolute error; MASE below 1: better than the seasonal naive method in sample
- Correlogram and Ljung–Box
  - $\hat{\gamma}(h)$: the sample autocovariance at lag h; $\hat{\rho}(h)$: the sample autocorrelation; T: the sample size
  - $Q^*(m)$: the sum of the first m squared autocorrelations, weighted; k: the number of estimated ARMA parameters
  - a Q^* above the $\chi^2(m - k)$ critical value (a small p-value) rejects white noise
- Models and criteria
  - ϕ: the AR(1) coefficient; θ: the MA(1) coefficient; $\ln L$: the maximised log-likelihood; k in AIC and BIC: all estimated parameters
  - ψ_j: the weights of the MA(∞) form of the model ($\psi_0 = 1$); σ: the standard deviation of the shocks; h: the horizon

Prerequisites for Today (3/22): the PACF and model identification

- **PACF** at lag h: the last coefficient ϕ_{hh} of an AR(h) regression

$$y_t = \phi_{h\mathbf{1}} y_{t-\mathbf{1}} + \cdots + \phi_{hh} y_{t-h} + e_t$$

  - the correlation of y_t and y_{t-h} after removing the effect of the lags in between
  - an AR(p) has $\phi_{hh} = 0$ for $h > p$: the PACF cuts off after lag p

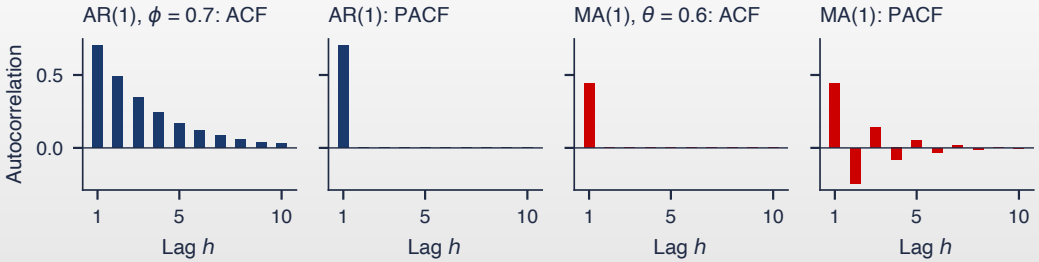
- **Ljung–Box** $Q^*(m)$: are the first m autocorrelations jointly zero?
  - H_0: white noise; p-value $= \Pr(\chi^2(m-k) > Q^*)$, with $m-k$ degrees of freedom on the residuals
  - a p-value below 0.05: the residuals still contain autocorrelation, change the model

- **Information criteria**: AIC and BIC of the formula sheet; for short samples, AICc

$$\text{AICc} = \text{AIC} + \frac{2k(k+1)}{T-k-1}$$

  - BIC penalises each extra parameter by $\ln T$ instead of 2

Prerequisites for Today (4/22): ACF and PACF in a chart



- AR(1): the ACF ϕ^h decays, the PACF cuts off after lag 1; MA(1): the ACF cuts off after lag 1, the PACF decays with alternating signs
- MA(1) here: $\rho(1) = 0.6/(1 + 0.36) = 0.441$; with data, compare each bar with the band $\pm 1.96/\sqrt{T}$

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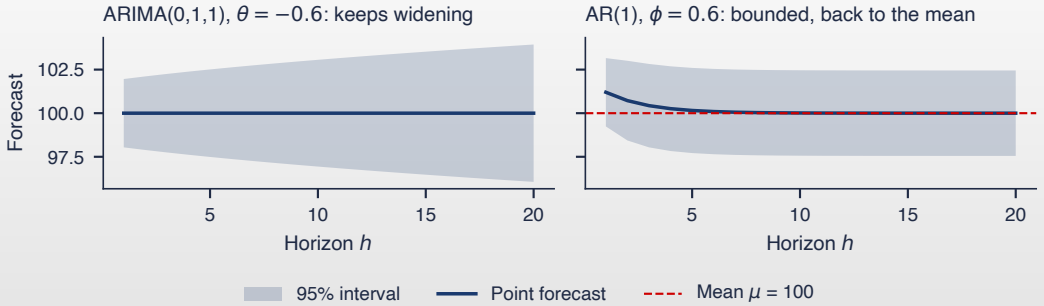
Prerequisites for Today (5/22): forecast intervals and the ψ weights

- Every ARMA or ARIMA model is a sum of past shocks, $y_t = \sum_{j \geq 0} \psi_j \varepsilon_{t-j}$; the **forecast error** at horizon h has variance

$$\mathrm{Var}(y_{T+h} - \hat{y}_{T+h}) = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2$$

  - only the shocks after the origin T are unknown: $\varepsilon_{T+1}, \ldots, \varepsilon_{T+h}$
- **AR(1)**: $\psi_j = \phi^j$
  - the sum converges to $1/(1-\phi^2)$: the interval stays bounded, the forecast returns to the mean
- **ARIMA(0,1,1)**: $\psi_0 = 1$, $\psi_j = 1 + \theta$ for $j \geq 1$
  - the sum $1 + (h-1)(1+\theta)^2$ grows linearly in h: the interval keeps widening (a unit root)
  - MA part invertible if $|\theta| < 1$: the root $-1/\theta$ of $1 + \theta z$ lies outside the unit circle

Prerequisites for Today (6/22): forecast intervals in a chart



- Left: ARIMA(0,1,1) with $\theta = -0.6$, $\sigma = 1$: a flat forecast, half-width $1.96\sqrt{1 + 0.16(h-1)}$, without limit
- Right: AR(1) with $\phi = 0.6$: the forecast returns to μ, the half-width tends to $1.96/\sqrt{1-0.36} = 2.45$

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Prerequisites for Today (7/22): unit roots, ARIMA, SARIMA, evaluation

Quantity	Formula
ADF	$\Delta y_t = c + bt + \gamma y_{t-1} + \sum_j \delta_j \Delta y_{t-j} + \varepsilon_t$; H_0: $\gamma = 0$ (unit root)
ADF, 5% critical values	about -2.87 (constant), -3.43 (constant and trend); reject if τ is smaller
KPSS	H_0: stationarity; 5% critical values 0.463 (constant), 0.146 (trend); reject if larger
ARIMA(0,1,1)	$\Delta y_t = \varepsilon_t + \theta \varepsilon_{t-1}$; $\hat{y}_{T+h} = y_T + \theta \varepsilon_T$, $\alpha_{SES} = 1 + \theta$
Its error variance	$\sigma^2[1 + (h-1)(1+\theta)^2]$
AR(1)×SAR(1)$_s$	$z_t = \phi z_{t-1} + \Phi z_{t-s} - \phi\Phi z_{t-s-1} + \varepsilon_t$
Annual rate from z	$z_t = \Delta\Delta_s \ln y_t$: $\Delta_s \ln y_{T+1} = \Delta_s \ln y_T + \hat{z}_{T+1}$
Diebold–Mariano	$d_t = e_{1t}^2 - e_{2t}^2$, $\mathrm{DM} = \bar{d}/(s_d/\sqrt{n})$; reject equal accuracy if $\lvert\mathrm{DM}\rvert > t_{0.975}(n-1)$

Prerequisites for Today (8/22): the notation of the previous sheet

- ADF and KPSS
  - $\Delta y_t = y_t - y_{t-\mathbf{1}}$; c: the constant; bt: the deterministic trend; δ_j: the coefficients of the lagged differences
  - $\gamma = 0$: a unit root; τ: the t-statistic of $\hat{\gamma}$, compared with Dickey–Fuller critical values, not Student ones
- ARIMA and SARIMA
  - ε_T: the last shock; α_{SES}: the weight of the equivalent exponential smoothing
  - s: the season length; Φ: the seasonal AR coefficient; $\Delta_s y_t = y_t - y_{t-s}$; z_t: the doubly differenced series
- Diebold–Mariano
  - $e_{\mathbf{1}t}$, $e_{\mathbf{2}t}$: the errors of the two forecasts; d_t: the difference of their squares; $\bar{d}$, s_d: its mean and standard deviation; n: the number of forecasts
  - $t_{\mathbf{0.975}}(n-1)$: the 97.5% quantile of the Student distribution with $n-1$ degrees of freedom

Prerequisites for Today (9/22): unit roots: ADF and KPSS

- **ADF** Dickey and Fuller (1979): subtract y_{t-1} from an AR(1) $y_t = \phi y_{t-1} + \varepsilon_t$

$$\Delta y_t = \gamma\, y_{t-1} + \varepsilon_t, \qquad \gamma = \phi - 1$$

  - $\gamma = 0$ ($\phi = 1$): a random walk, shocks are permanent; $\gamma < 0$: stationary, shocks die out
  - the lagged differences Δy_{t-j} of the sheet absorb the remaining autocorrelation

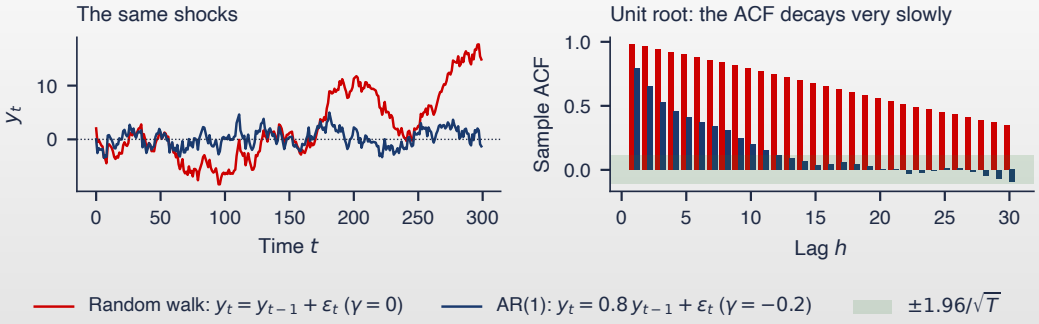
- **KPSS** Kwiatkowski et al. (1992): the opposite null, H_0: stationarity

$$\text{KPSS} = \frac{1}{T^2 \hat{\sigma}^2_{LR}} \sum_{t=1}^{T} S_t^2, \qquad S_t = \sum_{s=1}^{t} \hat{e}_s$$

  - $\hat{e}_s$: the residuals of y on a constant (or a trend); S_t: their partial sums; $\hat{\sigma}^2_{LR}$: the long-run variance of $\hat{e}$
  - a unit root makes S_t wander far from 0: KPSS large, reject stationarity

- Together: ADF does not reject and KPSS rejects: $I(1)$; ADF rejects and KPSS does not: $I(0)$; otherwise the evidence is mixed

Prerequisites for Today (10/22): a unit root in a chart



- Left: the same shocks; the random walk wanders without a fixed mean, the AR(1) with $\phi = 0.8$ keeps returning to 0
- Right: the sample ACF of the random walk stays high for 30 lags; $I(1)$: integrated of order 1, stationary after one difference

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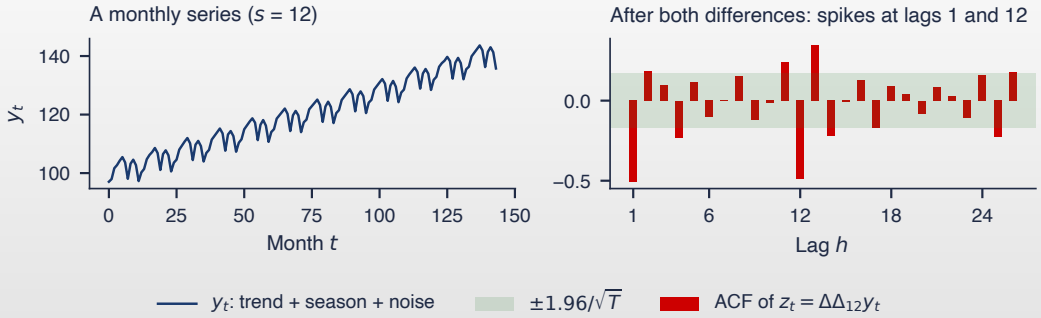
Prerequisites for Today (11/22): seasonal ARIMA and the airline model

- **Seasonal difference** $\Delta_s y_t = (1 - L^s) y_t = y_t - y_{t-s}$; $s = 12$ (monthly), $s = 4$ (quarterly)
  - $z_t = \Delta\Delta_s y_t$ removes the trend and a seasonal pattern that changes slowly
- **SARIMA**$(p, d, q)(P, D, Q)_s$ Box, Jenkins and Reinsel (2008): AR and MA polynomials in L and in L^s
  - the **airline model** $(0, 1, 1)(0, 1, 1)_{\mathbf{12}}$:

$$(1 - L)(1 - L^{\mathbf{12}})\, y_t = (1 + \theta L)(1 + \Theta L^{\mathbf{12}})\, \varepsilon_t$$

  - θ: the MA coefficient at lag 1; Θ: the seasonal MA coefficient at lag 12; z_t has ACF spikes at lags 1, 11, 12 and 13 only
- **Out-of-sample accuracy** on N test months
  - RMSE $= \sqrt{\frac{1}{N}\sum_i e_i^2}$, MAE $= \frac{1}{N}\sum_i |e_i|$; the seasonal naive forecast $\hat{y}_{T+h} = y_{T+h-\mathbf{12}}$ is the benchmark

Prerequisites for Today (12/22): seasonal differencing in a chart



- Left: a simulated monthly series with a trend and a seasonal pattern
- Right: the ACF of $z_t = \Delta\mathbf{\Delta_{12}} y_t$ of an airline model: negative spikes at lags 1 and 12, as $\theta < 0$ and $\Theta < 0$

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Prerequisites for Today (13/22): volatility and several series

Quantity	**Formula**
GARCH(1,1)	$\sigma^2_{t+1} = \omega + \alpha\varepsilon^2_t + \beta\sigma^2_t$, $\bar{\sigma}^2 = \omega/(1-\alpha-\beta)$, $h_{1/2} = \ln 0.5/\ln(\alpha+\beta)$
Annual volatility	$\sqrt{A\,\bar{\sigma}^2}$, $A =$ trading days a year
VaR 1%	$-(\mu + \sigma_{t+1} q_{0.01}(z))$; Student-$t_\nu$: $q = t_\nu^{-1}(0.01)\sqrt{(\nu-2)/\nu}$; Normal: -2.326
VAR(p)	$\mathbf{y}_t = \mathbf{c} + \mathbf{A}_1\mathbf{y}_{t-1} + \cdots + \mathbf{A}_p\mathbf{y}_{t-p} + \mathbf{u}_t$, OLS equation by equation
Granger F	$[(RSS_R - RSS_U)/p] \,/\, [RSS_U/(T-Kp-1)] \sim F(p, T-Kp-1)$
Long-run effect	$\partial y/\partial x = \sum_j b_{yx,j} \,/\, (1 - \sum_j b_{yy,j})$
Engle–Granger, ECM	ADF on OLS residuals, critical value about -3.34 (two variables); $h_{1/2} = \ln 0.5/\ln(1+\gamma)$
Johansen	trace $-T\sum_{i>r}\ln(1-\hat{\lambda}_i)$; test $r = 0, 1, \ldots$ in turn; stop at the first non-rejection

Prerequisites for Today (14/22): the notation of the previous sheet

- GARCH and VaR
  - $\sigma_t^{\mathbf{2}}$: the conditional variance; ε_t: the shock; ω: the constant; α: the reaction to shocks; β: the persistence
  - $\bar{\sigma}^{\mathbf{2}}$: the long-run variance; $h_{\mathbf{1/2}}$: the half-life of a volatility shock; μ: the mean return
  - $q_{\mathbf{0.01}}(z)$: the 1% quantile of the standardised innovations; ν: the degrees of freedom of the Student-t distribution
- VAR and Granger
  - $\mathbf{y}_t$: the vector of K variables; $\mathbf{c}$: the constants; $\mathbf{A}_i$: the coefficient matrix of lag i; $\mathbf{u}_t$: the shocks; p: the number of lags
  - RSS_R, RSS_U: the residual sums of squares without and with the lags of the tested variable
  - $b_{yx,j}$: the coefficient of x_{t-j} in the equation of y; $b_{yy,j}$: that of y_{t-j}
- Cointegration
  - $\gamma < 0$: the error-correction coefficient, the speed of return to equilibrium
  - $\hat{\lambda}_i$: the estimated eigenvalues of the Johansen procedure; r: the tested cointegration rank

Prerequisites for Today (15/22): GARCH: volatility clustering and the half-life

- **GARCH(1,1)** Bollerslev (1986): today's shock and yesterday's variance give tomorrow's variance
  - $\alpha + \beta < 1$: after a shock the variance returns to $\bar{\sigma}^2$
- **Variance forecast** h days ahead

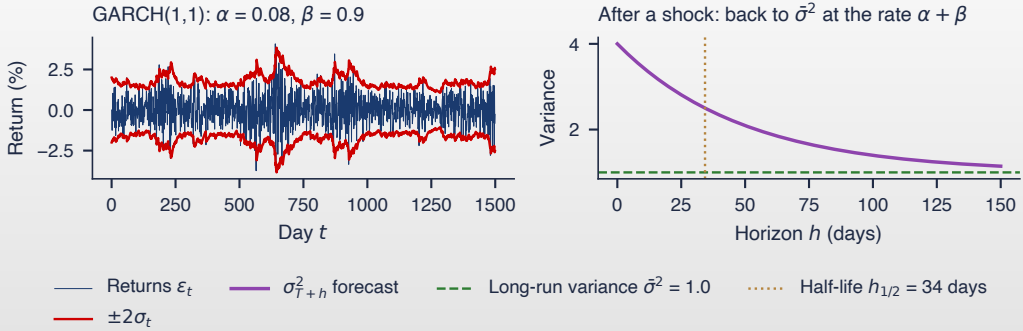
$$\sigma^2_{T+h} - \bar{\sigma}^2 = (\alpha + \beta)^{h-1} (\sigma^2_{T+1} - \bar{\sigma}^2)$$

  - the half-life $h_{1/2} = \ln 0.5 / \ln(\alpha + \beta)$ solves $(\alpha + \beta)^h = 0.5$: the gap to $\bar{\sigma}^2$ halves
- **ARCH-LM test** Engle (1982) with q lags: regress $\hat{\varepsilon}^2_t$ on $1, \hat{\varepsilon}^2_{t-1}, \ldots, \hat{\varepsilon}^2_{t-q}$

$$\text{LM} = T\, R^2 \sim \chi^2(q) \text{ under } H_0\text{: no ARCH effects}$$

  - $\hat{\varepsilon}_t$: the demeaned returns; R^2: of this regression; a large LM: the size of returns is predictable

Prerequisites for Today (16/22): GARCH in a chart



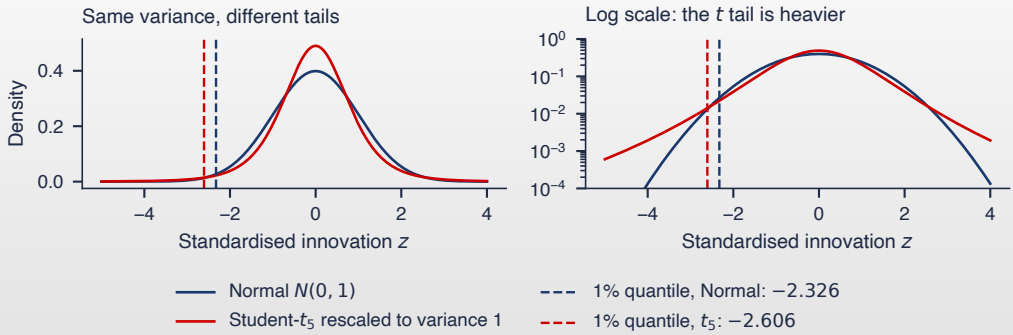
- Left: simulated GARCH(1,1) returns; calm and turbulent periods alternate (volatility clustering), $\pm 2\sigma_t$ follows them
- Right: $\alpha + \beta = 0.98$, so $h_{1/2} = \ln 0.5 / \ln 0.98 = 34$ days: a variance four times $\bar{\sigma}^2$ halves its excess in 34 days

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Prerequisites for Today (17/22): VaR with Normal and Student-t innovations

- Return $r_{t+\mathbf{1}} = \mu + \sigma_{t+\mathbf{1}} z_{t+\mathbf{1}}$, with z of mean 0 and variance 1; **VaR 1%**: the loss exceeded with probability 1%
  - VaR 1% $= -(\mu + \sigma_{t+\mathbf{1}} q_{\mathbf{0.01}}(z))$ (formula sheet) is reported as a positive loss, at the level "1%" (the tail probability)
- **Student-t_ν** innovations: t_ν has variance $\nu/(\nu-2)$, so it is rescaled to variance 1
  - hence the factor $\sqrt{(\nu-2)/\nu}$ in $q_{\mathbf{0.01}}(z)$ of the formula sheet; $t_\nu^{-\mathbf{1}}(0.01)$: the 1% quantile of the Student distribution with ν degrees of freedom; small ν: heavy tails

Prerequisites for Today (18/22): VaR in a chart



- Normal and Student-t_5 with the same variance: the t has more mass in the centre and in the tails
- 1% quantiles: -2.326 (Normal) and $-3.365 \cdot \sqrt{3/5} = -2.606$ (t_5): with $\mu = 0$, $\sigma = 1\%$, VaR 1% is 2.33% or 2.61%

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Prerequisites for Today (19/22): Granger causality and cointegration

- **Granger causality** Granger (1969): x Granger-causes y if the lags of x improve the forecast of y, given the lags of y
  - H_0: all $b_{yx,j} = 0$, tested with the F statistic of the sheet; extra predictability, not proof of a causal mechanism
- **Cointegration** Engle and Granger (1987): y_t, x_t are $I(1)$, but a combination is stationary

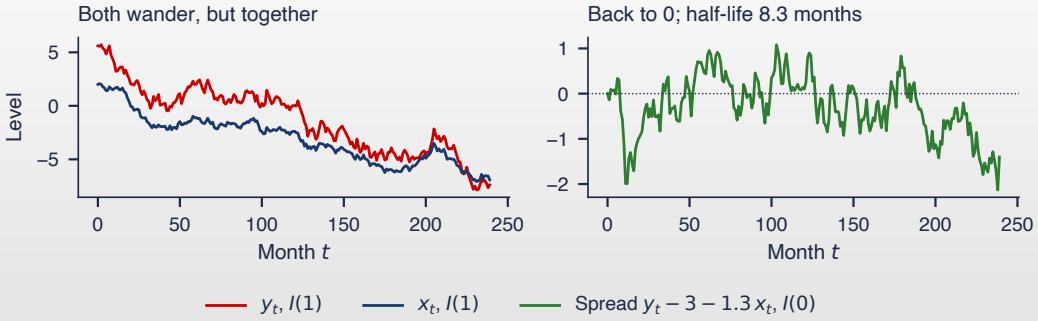
$$u_t = y_t - c - \beta x_t \sim I(0), \qquad \Delta y_t = \gamma\, u_{t-\mathbf{1}} + \delta\, \Delta x_t + \varepsilon_t$$

  - β: the long-run coefficient; the second equation is the **error-correction model** (ECM); δ: the short-run effect
- **Johansen** Johansen (1991): eigenvalues $\hat{\lambda}_\mathbf{1} > \hat{\lambda}_\mathbf{2} > \dots$ of a VECM; two statistics for rank r

$$\lambda_{trace}(r) = -T \sum_{i>r} \ln(1 - \hat{\lambda}_i), \qquad \lambda_{max}(r) = -T \ln(1 - \hat{\lambda}_{r+\mathbf{1}})$$

  - large values reject "at most r cointegrating relations"; trace: against up to K relations; maximum eigenvalue: against $r+1$

Prerequisites for Today (20/22): cointegration in a chart



- Left: two simulated $I(1)$ series that wander together; neither has a fixed mean
- Right: the spread $u_t = y_t - 3 - 1.3x_t$ is stationary; with $\gamma = -0.08$, $h_{1/2} = \ln 0.5 / \ln 0.92 = 8.3$ months

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Prerequisites for Today (21/22): extensions

Quantity	Formula
ARFIMA$(0, d, 0)$	$\rho(1) = d/(1-d)$, $H = d + 1/2$; stationary for $d < 1/2$, mean-reverting for $d < 1$
Local level	$F_t = P_t + \sigma_\varepsilon^2$, $K_t = P_t/F_t$, $a_{t\|t} = a_t + K_t(y_t - a_t)$, $P_{t+1} = P_t(1-K_t) + \sigma_\eta^2$
Steady state	$\bar{P}/\sigma_\varepsilon^2 = (q + \sqrt{q^2+4q})/2$, $\bar{K} = \alpha_{SES}$, $q = \sigma_\eta^2/\sigma_\varepsilon^2$
Markov switching	$E(D_i) = 1/(1-p_{ii})$, $\pi_1 = (1-p_{22})/(2-p_{11}-p_{22})$
Machine learning	walk-forward validation only; compare with a simple benchmark
Values	$\chi^2_{0.95}$: 3.84 (1), 5.99 (2), 9.49 (4), 12.59 (6), 18.31 (10); $F_{0.95}(2, 250) = 3.03$

Prerequisites for Today (22/22): the notation of the previous sheet

- Long memory
  - d: the order of fractional integration; H: the Hurst exponent; $\rho(1)$: the first autocorrelation
- Local level model and the Kalman filter
  - a_t, P_t: the predicted level and its variance; $a_{t|t}$: the level updated with y_t
  - $\sigma_{\varepsilon}^{\mathbf{2}}$, $\sigma_{\eta}^{\mathbf{2}}$: the variances of the measurement and level noise; q: their ratio
  - F_t: the variance of the prediction error $y_t - a_t$; K_t: the Kalman gain, the share of the surprise added to the level; $\bar{P}$, $\bar{K}$: their steady-state values
- Markov switching and critical values
  - p_{ii}: the probability of staying in regime i; $E(D_i)$: the expected duration of regime i; $\pi_{\mathbf{1}}$: the long-run share of regime 1
  - $\chi^{\mathbf{2}}_{\mathbf{0.95}}$: 95% quantiles of the chi-square distribution, degrees of freedom in brackets; $F_{\mathbf{0.95}}(2, 250)$: the 95% quantile of the F distribution

A1: an ARIMA(0,1,1) by hand [Solved]

Task

- $\Delta y_t = \varepsilon_t - 0.6\,\varepsilon_{t-1}$, $\varepsilon_t \sim WN(0,1)$; y_0 given, $\varepsilon_0 = 0$; today $y_T = 100$, $\hat{\varepsilon}_T = 1.5$.
- 1. Show that Δy_t is stationary and invertible, and compute $\rho_{\Delta y}(1)$.
- 2. Compute $\mathrm{Var}(y_t)$ for $t = 10$ and $t = 100$ and say whether y_t is stationary.
- 3. Forecast y_{T+1} and y_{T+4}; give the 95% interval for $h = 4$.
- 4. Give the weight of the equivalent exponential smoothing.
- Report: five numbers, an interval and one sentence.

Solution

- 1. An MA(1) is always stationary; root $1/0.6 = 1.667 > 1$: invertible; $\rho(1) = -0.6/1.36 = -0.441$
- 2. $y_t = y_0 + \varepsilon_t + 0.4\sum_{j<t}\varepsilon_j$: $\mathrm{Var} = 1 + (t-1)0.16$: 2.44 and 16.84: grows with t, not stationary
- 3. $\hat{y}_{T+h} = 100 - 0.6 \times 1.5 = 99.10$ for every h; variances 1.00 and 1.48; interval $[96.72, 101.48]$
- 4. $\alpha = 1 + \theta = 0.4$
- Interpretation: the forecast is flat, like that of exponential smoothing, but its interval keeps widening: a unit root

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A2: identifying the Romanian unemployment rate [Proposed]

Solution discussed in class

Task

Series	ADF τ	p	ADF crit. 5%	KPSS (crit. 0.463)
u_t	-1.52	0.524	-2.87	1.037
Δu_t	-8.12	< 0.001	-2.87	0.096

- Unemployment rate u_t (seasonally adjusted, %), January 2005 – August 2026; both tests with a constant. Δu_t ($T = 259$): ACF -0.19, 0.11, 0.09, -0.07; PACF -0.19, 0.08, 0.13, -0.04 (lags 1–4). Model: Seminar 3, A1; Seminar 2, A6.
- 1. State H_0 of each test and decide the order of integration of u_t.
- 2. Compute the band $\pm 1.96/\sqrt{T}$ and Ljung–Box $Q(4)$ from the four ACF values; decide at 5%.
- 3. Propose an ARIMA model for u_t.
- Report: two decisions, two numbers, a model.

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A3: SARIMA output for Romanian GDP [Proposed]

Solution discussed in class

Task

Dependent: z_t	Coef.	SE	z	p
AR(1)	−0.083	0.106	−0.79	0.431
SAR(4)	−0.405	0.085	−4.79	< 0.001
SIGMASQ	10.41			
Ljung–Box $Q(8)$	18.14			0.006

- $z_t = \Delta\Delta_4\, 100 \ln \mathrm{GDP}_t$, real GDP not seasonally adjusted, 2001Q2 – 2026Q2 ($T = 101$). Last values of z: 1.923 (2025Q2), −1.150 (2025Q3), −2.584 (2025Q4), −0.247 (2026Q1), −0.457 (2026Q2); annual growth $\Delta_4\, 100 \ln \mathrm{GDP}$ in 2026Q2: −1.96%. Model: Seminar 4, A4.
- 1. Write the model equation and say which coefficients are significant.
- 2. Test whether the residuals are white noise ($\chi^2_{0.95}(6) = 12.59$).
- 3. Forecast z and the annual growth rate for 2026Q3.

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A4: VAR output and Granger causality [Proposed]

Solution discussed in class

Task

	π_t equation	i_t equation
π_{t-1}	1.318 [21.58]	0.031 [0.48]
π_{t-2}	−0.342 [−5.52]	0.008 [0.13]
i_{t-1}	0.011 [0.19]	1.167 [19.60]
i_{t-2}	−0.012 [−0.22]	−0.219 [−3.81]
const.	0.112 [1.62]	0.063 [0.88]
R^2	0.97	0.97

- VAR(2) for Romanian annual inflation π and ROBOR 3M i (%), March 2005 – August 2026, $T = 258$; t in brackets. Without the lags of π, the RSS of the i equation rises from 89.81 to 92.20. Model: Seminar 6, A5.
- 1. Write the i equation.
- 2. Compute the Granger F statistic for "π does not Granger-cause i" and decide at 5% ($F_{0.95} = 3.03$).
- 3. Compute the long-run effect on ROBOR of a permanent 1 pp rise in inflation.

A5: GARCH output for the DAX and VaR 1% [Proposed]

Solution discussed in class

Task

DAX	μ	ω	α	β	ν
estimate	0.081	0.0206	0.097	0.895	7.08
SE	0.012	0.0049	0.011	0.012	0.617

- GARCH(1,1)-t, daily DAX log returns in % since 2000 ($n = 6\,786$); last day (September 2026): $\sigma^2_T = 0.662$, $\hat{\varepsilon}_T = -1.698$; about 254 trading days a year; use $\nu = 7$: $t_7^{-1}(0.01) = -2.998$. Model: Seminar 5, A1 and A4.
- 1. Compute the persistence, the half-life and the long-run annual volatility.
- 2. Compute σ^2_{T+1}.
- 3. Compute the one-day VaR 1% and compare it with the Normal VaR 1%.

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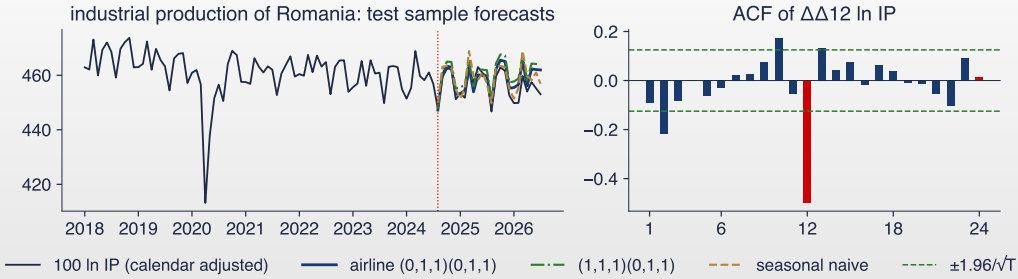
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B1: Box–Jenkins for Romanian industrial production [Solved]

- **Question**: which SARIMA model forecasts Romanian industrial production best, and does it beat the seasonal naive method?
- **Inputs**: industrial production (B–D), calendar adjusted, monthly, January 2005 – July 2026; test sample: the last 24 months
- **Tasks**
  1. Run ADF and KPSS on $100 \ln y_t$, its first difference and $z_t = \Delta\Delta_{\mathbf{12}} 100 \ln y_t$.
  2. Plot the ACF of z_t and propose a model.
  3. Fit the airline model and SARIMA$(1,1,1)(0,1,1)_{\mathbf{12}}$ on the training sample; compare AICc, BIC and Ljung–Box.
  4. Forecast the 24 test months with both models and with the seasonal naive method; compare RMSE and MAE.
  5. Interpretation: which model would you use to forecast, and why?
- **Report**: a table of tests, a table of two models, three RMSE and one sentence
- Notebook: section B1

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B1: solution (1/2) [Solved]



Model	AICc	BIC	LB(12) p	LB(24) p	test RMSE
airline $(0,1,1)(0,1,1)_{12}$	1304.2	1314.3	0.065	0.432	3.75
$(1,1,1)(0,1,1)_{12}$	1291.1	1304.5	0.789	0.972	5.29
seasonal naive					3.85

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B1: solution (2/2) [Solved]

- Tests: ADF $\tau = -1.31$ ($p = 0.885$) and KPSS $0.523 > 0.146$ for the level; z_t: ADF -6.47, KPSS 0.055: $d = D = 1$; ACF of z_t: -0.09, -0.22 at lags 1, 2, -0.50 at 12
- Airline: $\hat{\theta} = -0.205$, $\hat{\Theta} = -0.934$; the $(1, 1, 1)$ model: $\hat{\phi} = 0.574$, $\hat{\theta} = -0.828$, almost cancelling factors
- Interpretation: the $(1, 1, 1)$ model wins in sample (BIC, clean residuals) but loses out of sample; the airline model is slightly better than the seasonal naive method: use the airline model, and keep the seasonal naive method as the benchmark

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B2: volatility of the BET since 2015 [Proposed]

- **Question**: is the Bucharest market calmer or more turbulent today than usual, and what does it mean for tomorrow's VaR 1%? Model: Seminar 5, B1.
- **Inputs**: BET daily log returns in %, January 2015 – September 2026 ($n = 2\,925$)
- **Tasks**
  1. Run the ARCH-LM test with five lags on the demeaned returns.
  2. Fit a GARCH(1,1) with Student-t innovations; report the persistence and the half-life.
  3. Compare the long-run, the sample and today's annualised volatility.
  4. Compute tomorrow's VaR 1% with the t and with the Normal quantile.
  5. Interpretation: is the BET calmer or more turbulent than usual today?
- **Report**: one test, five parameters, three volatilities, two VaR values and one sentence
- Notebook: section B2

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B3: the Romanian and German 10-year yields [Proposed]

- **Question**: is the Romanian 10-year yield tied to the German one in the long run? Model: Seminar 7, B1 and A3.
- **Inputs**: monthly means of daily 10-year yields (%), January 2008 – September 2026 ($T = 224$)
- **Tasks**
  1. Run ADF on both yields and on their differences.
  2. Run the Engle–Granger test (Romania on Germany) and report $\hat{\beta}$.
  3. Run the Johansen trace and maximum-eigenvalue tests.
  4. Estimate the error-correction equation of each yield and compute the half-life.
  5. Interpretation: is the Romanian yield tied to the German one?
- **Report**: four ADF results, two cointegration tests, two adjustment coefficients and one sentence
- Notebook: section B3

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C1: a forecast competition for Romanian inflation [Proposed]

- **Question**: which method would have forecast Romanian monthly inflation best over the last ten years, one and twelve months ahead?
- **Inputs**: the Romanian HICP (Eurostat), monthly, 2005–2026; Lecture 15 runs one SARIMA on one test sample
- **Tasks**
  1. List five competitors: seasonal naive, SES, SARIMA, ARFIMA, a ridge model with lags and month dummies.
  2. Describe a rolling-origin design: first origin, step, horizons, refits.
  3. Say how you would treat the tax changes of 2010, 2015 and 2025.
  4. Choose the error measures and the test for the comparison.
- **Report**: a one-page plan that a team could turn into its project
- Notebook: section C1

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C2: audit an AI answer [Proposed]

- A student asked an AI assistant to check the answers to Part A. The answer:
  - `(a) For the unemployment rate, ADF gives p = 0.52, so the unit root is rejected at 5%.`
  - `(b) KPSS = 1.04 > 0.463 rejects, so the level of unemployment is stationary.`
  - `(c) For the DAX, alpha + beta = 0.992, so a shock halves in 1/(1 - 0.992) = 128 days.`
  - `(d) The VaR 99% of the DAX for tomorrow is -2.31%.`
  - `(e) Inflation Granger-causes ROBOR (p = 0.036), which proves that inflation causes the central bank to raise rates.`
  - `(f) Ljung-Box Q(12) on the residuals of an AR(1) x SAR(1) model has 12 degrees of freedom.`
- **Tasks**
  1. For each statement, say whether it is correct; if not, give the correct statement and the correct number (notebook, section C2).
  2. Report: six verdicts with one line of justification each.

Key takeaways

- Every exam problem has three parts: the formula with the numbers, the result with its unit and sign, the interpretation
- Tests: state H_0 first; ADF and KPSS have opposite nulls; Ljung–Box on residuals loses k degrees of freedom
- Forecasts by hand: write the model in levels of z_t, substitute the last values, then undo the differences
- A better in-sample model is not a better forecaster (B1); judge it on a test sample against a benchmark
- An AI answer is a draft: check the null hypotheses, the half-life, the VaR level and the degrees of freedom

After the seminar

- Lecture 15 gathers the course: the course map, two recap slides per chapter, the toolbox, Box–Jenkins on Romanian inflation, eight solved exam-type problems, the project
- Try the [Proposed] tasks on paper first, then check them in the notebook
- C1 can grow into a team project
- More practice: the practice set of 25 exam problems with numerical answers
- Reading: Huang and Petukhina (2022); Hyndman and Athanasopoulos (2021); Box, Jenkins and Reinsel (2008)

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